Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Distance and midpoint formulas

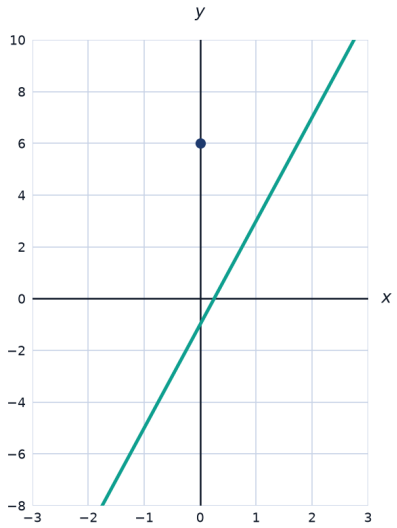
Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.

2 [Foundational] • Target: Slope and equations of lines

Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.

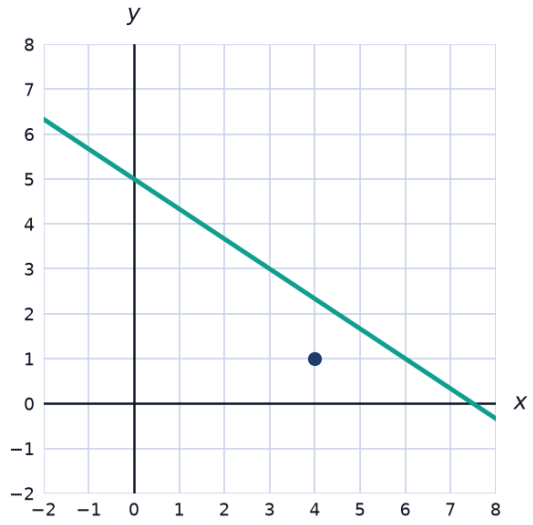
3 [Foundational] • Target: Parallel and perpendicular lines

Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.



4 [Foundational] • Target: Parallel and perpendicular lines

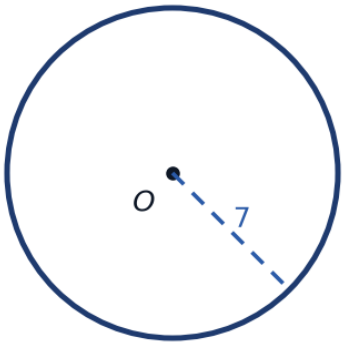
Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.



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5 [Foundational] • Target: The equation of a circle

Write the equation of a circle with center $(3, -2)$ and radius 7.



6 [Foundational] • Target: The equation of a circle

Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, by completing the square.

7 [Foundational] • Target: Reflections, translations, and symmetry

Find the coordinates of the point $(5, -3)$ after a reflection over the x-axis, then after a reflection over the y-axis.

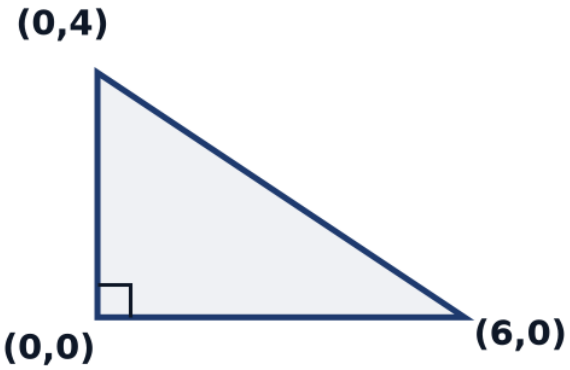
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Reflections, translations, and symmetry

A point (a, b) is translated 3 units right and 5 units down. Write the coordinates of the image point.

9 [Intermediate] • Target: Finding area using the coordinate plane

Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.



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10 [Intermediate] • Target: Finding area using the coordinate plane

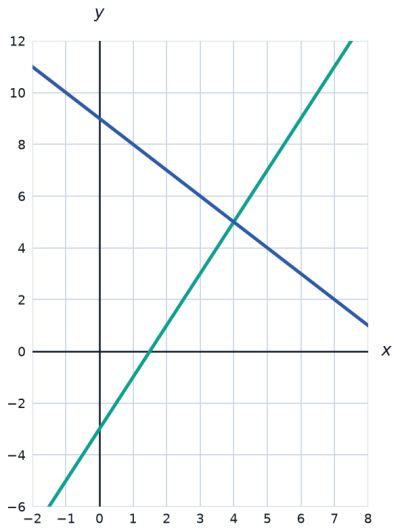
A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.

11 [Intermediate] • Target: Determining if points are collinear

Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.

12 [Intermediate] • Target: Finding the intersection of two lines

Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.



13 [Intermediate] • Target: Combining coordinate geometry concepts

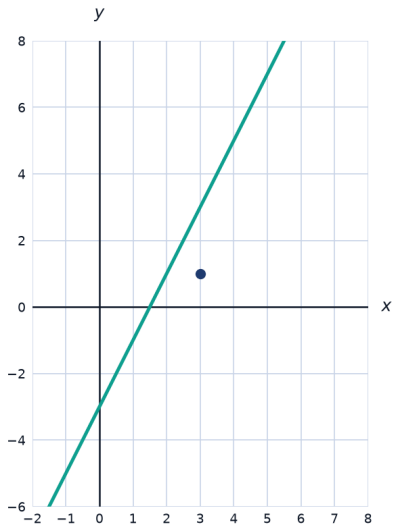
This question has two parts. Triangle ABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.



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14 [Intermediate] • Target: Graphing inequalities in the coordinate plane

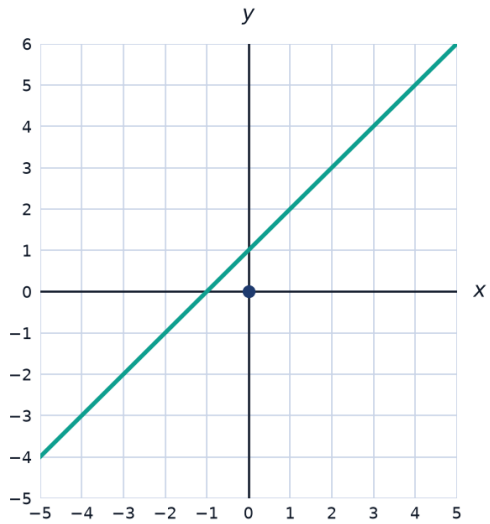
Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 5.01 Plane Geometry

A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.



16 [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 6.01 Trigonometry

Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.

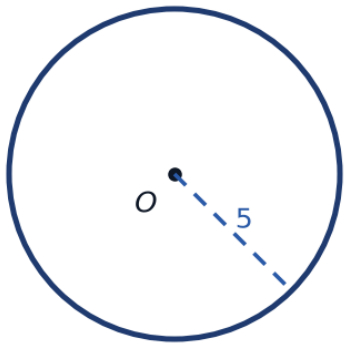
17 [Challenging] • Target: Finding points that divide a segment in a given ratio • Cross-Lesson: 7.01 Statistics & Probability

Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1:3$ (closer to the first point).

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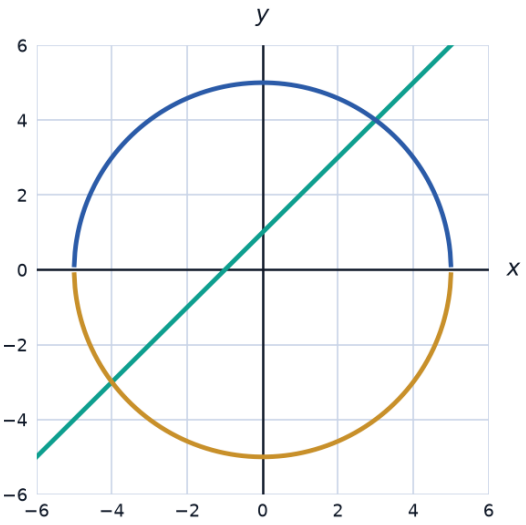
18 [Challenging] • Target: A multi-part problem combining lines and circles • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

This question has two parts. A circle has equation $(x-2)^2+(y+1)^2=25$. (a) State the center and radius. (b) Determine whether the point $(5,3)$ lies inside, on, or outside the circle, by computing its distance from the center.



19 [Challenging] • Target: Systems of a line and a circle • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the points of intersection of the line $y=x+1$ and the circle $x^2+y^2=25$.



20 [Challenging] • Target: Finding area using the coordinate plane • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Point $M(3,\ -2)$ is the midpoint of line segment $\overline{AB}$. If point A has coordinates $(-1,4)$, what are the coordinates of point B?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Distance and midpoint formulas

Find the radius of the circle $x^2 + y^2 + 16x - 18y + 81 = 0$, by completing the square.

a. 8

b. 7

c. 64

d. 9

22 [Foundational] • Target: Slope and equations of lines

Find the equation of the line perpendicular to $y = -\frac{5}{4}x$ passing through $(0, -3)$, in slope-intercept form.

a. $y = -\frac{4}{5}x - 3$

b. $y = \frac{4}{5}x - 3$

c. $y = -\frac{5}{4}x - 3$

d. $y = \frac{4}{5}x - 2$

23 [Foundational] • Target: Parallel and perpendicular lines

In the standard (x, y) coordinate plane, line l contains the points $(2, 5)$ and $(6, -3)$. Line k is perpendicular to the segment connecting these two points and passes through its midpoint. What is the y-intercept of line k?

a. -6

b. -1

c. 3

d. 9

24 [Foundational] • Target: The equation of a circle

In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle $\angle AGH$ is 125°, what is the measure of angle $\angle GHD$?

Diagram description: Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G is on AB, H is on CD. Angle AGH is in the upper left quadrant formed by the intersection.

a. 55°

b. 125°

c. 90°

d. 180°

25 [Foundational] • Target: Reflections, translations, and symmetry

In the coordinate plane, the circle $x^2 + y^2 = 25$ intersects the line $y = 3$ at two points. What is the distance between these two points?

a. 6

b. 8

c. 10

d. 12

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26 [Foundational] • Target: Finding area using the coordinate plane

In ΔABC, the lengths of the sides are $AB = 13$, $BC = 14$, and $AC = 15$. A line segment $\overline{DE}$ is drawn parallel to $\overline{BC}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$. If the perimeter of ΔADE is equal to the perimeter of trapezoid $BCED$, what is the length of $\overline{DE}$?

a. 7

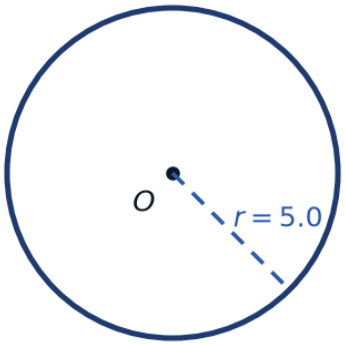
b. $\frac{42}{5}$

c. $\frac{21}{2}$

d. $7\sqrt{2}$

27 [Foundational] • Target: Determining if points are collinear

In the coordinate plane, the circle $(x - 2)^2 + (y - 3)^2 = 25$ intersects the y-axis at two points. What is the distance between these two points?



a. $2\sqrt{21}$

b. $4\sqrt{5}$

c. 8

d. 10

28 [Foundational] • Target: Finding the intersection of two lines

Two parallel lines are cut by a transversal. One angle measures 105°. What is its alternate exterior angle?

a. 168°

b. 67°

c. 91°

d. 105°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Combining coordinate geometry concepts

Two parallel lines are cut by a transversal. One angle measures 86°. What is its alternate interior angle?

a. 124°

b. 86°

c. 119°

d. 176°

30 [Intermediate] • Target: Graphing inequalities in the coordinate plane

Two parallel lines are cut by a transversal. One angle measures 130°. What is its co-interior (same-side interior) angle?

a. 12°

b. 108°

c. 100°

d. 50°

31 [Intermediate] • Target: The equation of a parabola in the coordinate plane

Two parallel lines are cut by a transversal. One angle measures 60°. What is its corresponding angle?

a. 60°

b. 74°

c. 144°

d. 118°

32 [Intermediate] • Target: Finding points that divide a segment in a given ratio

Two parallel lines are cut by a transversal. One angle measures 120°. What is its vertical angle?

a. 162°

b. 120°

c. 139°

d. 11°

33 [Intermediate] • Target: A multi-part problem combining lines and circles

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

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34 **[Intermediate]** • **Target: Systems of a line and a circle**

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?

a. -7

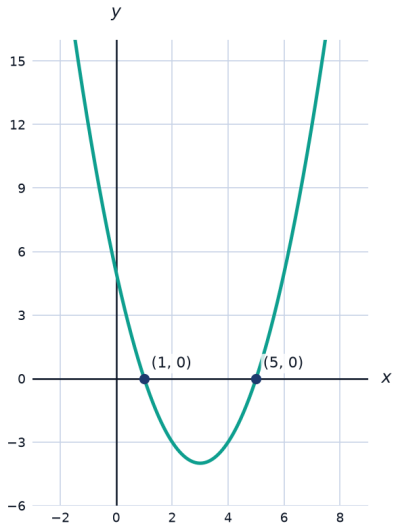
b. 9

c. 12

d. 15

35 **[Intermediate]** • **Target: Distance and midpoint formulas**

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



a. 12

b. 8

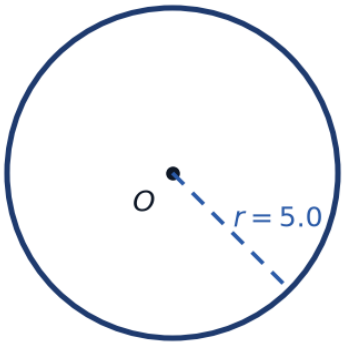
c. 4

d. -3

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36 [Intermediate] • Target: Slope and equations of lines

The line $x = 5$ intersects the circle $(x - 2)^2 + (y + 1)^2 = 25$ at two points. What is the distance between these two points?



a. 6

b. 8

c. 10

d. 12

37 [Intermediate] • Target: Parallel and perpendicular lines

A line has slope -2. What is the slope of a line perpendicular to it?

a. $\frac{1}{10}$

b. $\frac{2}{9}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The equation of a circle • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

The parabola $y = x^2 - x - 2$ crosses the x-axis at two points. What is the sum of their x-coordinates?

a. -4

b. 1

c. -2

d. 5

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39 [Challenging] • Target: Reflections, translations, and symmetry • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A line has slope 2. What is the slope of a line parallel to it?

a. $\frac{51}{100}$

b. 2

c. $\frac{13}{20}$

d. $\frac{1}{2}$

40 [Challenging] • Target: Finding area using the coordinate plane • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$

b. 4

c. $\frac{10}{3}$

d. $2\sqrt{6}$

41 [Challenging] • Target: Determining if points are collinear • Cross-Lesson: 3.01 Functions & Graphs

In the xy-plane, the points $(1, 3)$ and $(7, 11)$ are the endpoints of a diameter of a circle. What is the equation of the tangent line to the circle at the point $(7, 11)$?

a. $3x + 4y = 65$

b. $4x + 3y = 61$

c. $3x - 4y = -23$

d. $4x - 3y = -5$

42 [Challenging] • Target: Finding the intersection of two lines • Cross-Lesson: 5.01 Plane Geometry

The line $y = 3x - 5$ intersects the parabola $y = x^2 - 2x + 1$ at points P and Q. What is the x-coordinate of the midpoint of segment PQ?

a. 2

b. 2.5

c. 3

d. 3.5

43 [Challenging] • Target: Combining coordinate geometry concepts • Cross-Lesson: 6.01 Trigonometry

In the standard (x, y) coordinate plane, the vertices of ΔABC are $A(0, 0)$, $B(10, 0)$, and $C(2, k)$, where $k > 0$. If $\angle ACB = 90^\circ$, what is the value of k?

a. $2\sqrt{2}$

b. 4

c. $2\sqrt{5}$

d. 8

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44 [Challenging] • Target: Graphing inequalities in the coordinate plane • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation $z(2 - i) = 15 + 5i$ where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = [x; y]$. If matrix $A = [3, -1; 2, 4]$, what is the magnitude of the vector $A\mathbf{v}$?

a. $5\sqrt{2}$

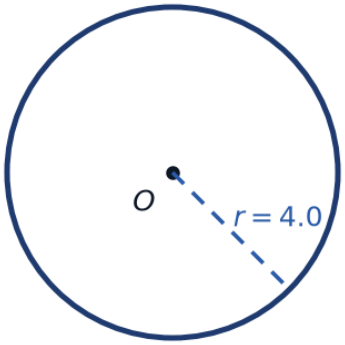
b. $10\sqrt{5}$

c. $10\sqrt{10}$

d. 40

45 [Challenging] • Target: The equation of a parabola in the coordinate plane • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

In the xy-plane, a line passes through the center of the circle $(x - 3)^2 + (y + 2)^2 = 16$ and is perpendicular to $2x - 4y = 9$. What is the y-intercept of this line?



a. $(0, 4)$

b. $(0, 2)$

c. $(0, -2)$

d. $(0, -8)$

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